

Karan Naidu

 [LinkedIn](#)

 +91-9425541122

 [Portfolio](#)

 naidu.karan10@gmail.com

 [GitHub](#)

 [LeetCode](#)

Summary

Backend and System Software Engineer with 3+ years of experience in both embedded and general-purpose systems. Experienced in building stable, efficient tools, from automotive audio features to network data processing. Focused on solving real-world problems with practical code and clear results.

Skills

- C++ | C | Python | Java | React | HTML | CSS | XML | SQL | MongoDB
- Git | GitLab | Jira | Perforce | Wireshark | IXIA | Niagara | QNX | OpenTelemetry
- Backend | Embedded | Networking | Observability | Debugging | Linux | System Programming

Experience

Engineer

Cleartrail Technologies

India

Feb 2023 – Present

Tech Stack: C, C++, Python, Shell Scripting, Linux, Windows, Jira, GitLab, Wireshark, Niagara, IXIA

- Planned and implemented optimization features across network pipeline components using C++, Shell scripting, and Python; collaborated with system teams to reduce packet redundancy and reduce latency by 40%, improving real-time throughput.
- Created support for GTP control packet analysis to extract Information Elements from mobile network data, accelerating diagnostics, and ensuring seamless integration with cross-functional teams.
- Built a Linux-based sanity testing tool in C++, increasing analysis accuracy by 50% and validation speed; resolved 500+ issues via Klocwork and Valgrind, enhancing runtime performance by 25% and code quality by 95%.

Engineer

Qualcomm

India

Jun 2021 – Sep 2022

Tech Stack: C, C++, Python, Java, Linux, Windows, Jira, GitLab, Perforce, QNX

- Contributed to the implementation of out-of-band memory management for new audio features in QNX / Linux embedded systems, helping to reduce interface latency by 30% and improve overall system efficiency.
- Implemented a synchronization solution in C to resolve clock drift in audio systems, improving playback stability by 20%; collaborated closely with the DSP team.

Intern

Qualcomm

India

Jan 2021 – Jun 2021

Tech Stack: C, Python, Linux, Jira, GitLab, Perforce

- Validated and Enhanced automotive audio systems, including Multi-Zone Audio, ANC, and Engine Sound Enhancement, by addressing platform-specific issues across QNX and Linux, boosting system reliability.

Intern

Qualcomm

India

May 2020 – Jul 2020

Tech Stack: C, Python, Linux, Jira, GitLab, Perforce

- Developed a Python/C-based configuration toolkit with a cross-platform GUI to visualize and modify DTB and DTS files; adopted by 5+ teams, improving hardware development workflows.

Intern

Continental

India

May 2019 – Aug 2019

Tech Stack: Python, GitHub

- Built and optimized Decision Tree and LSTM models for obstacle classification using RADAR sensor data; achieved 80% TPR with <4% error, enhancing object and path detection and multimodal perception in autonomous driving systems.

Projects

- **AI Agent for Mario:** Built a reinforcement learning agent to analyze platform layouts and optimize character movement for adaptive gameplay.
- **Encrypted Evaluation of Decision Trees:** Developed a privacy-preserving model evaluation system using homomorphic encryption for secure inference.

- **TextWorld Games:** Designed an NLP-driven RL agent to improve decision-making and contextual understanding in text-based environments.
- **Content Repurposement System using eXist-db:** Created a transformation pipeline using XQuery and eXist-db to convert multiformat data into structured XML, exported as HTML/PDF.

Education

Integrated Master of Technology	<i>IIT Bangalore, India</i>	08/2016 – 05/2021
Class 12 – 89.2%	<i>Christ Church Boys' Senior Secondary School, India</i>	05/2015